Supplemental Materials (SM), July 5, 2026

Appendix C: Identifiability in the Mixture Model

We have $(\underset{\sim}{x}_i, y_i), i = 1, \ldots, n$, where $\underset{\sim}{x}_i$ are covariates and y_i are values of the study variable, and we consider the mixture model,

$$y_i \mid p, \underset{\sim}{\beta}, \sigma^2, \rho \overset{ind}{\sim} (1-p)\text{Normal}\left(\underset{\sim}{x}_i'\underset{\sim}{\beta}, \rho\sigma^2\right) + p\text{Normal}\left(\underset{\sim}{x}_i'\underset{\sim}{\beta}, \sigma^2\right), 0 < p < \frac{1}{2}, 0 < \rho < 1.$$

Without loss of generality, we drop the survey weights. Note that for each $i = 1, \ldots, n$, the two components have the same mean, $\underset{\sim}{x}_i'\underset{\sim}{\beta}$, so that the distribution is symmetric about the mean, and outliers can be in both tails, although we can consider outliers only in the right tail. Here, we show that the parameters $p, \underset{\sim}{\beta}, \rho, \sigma^2$ are identifiable. We also show that the joint posterior density of $\pi(p, \underset{\sim}{\beta}, \rho, \sigma^2 \mid \underset{\sim}{y})$ under flat priors is proper.

Identifiability is demonstrated in the following steps.

1. Separate the data into two groups, G and G^c. This can be done by ordering the $y_1, \ldots, y_n$ from smallest to largest and looking for a separation among the larger values. Let $m > \frac{n}{2}$ denote the cardinality of G. Then $\hat{p} = (n-m)/n$.

2. Since the two components of the y_i have the same mean, we can use least squares to get $\hat{\underset{\sim}{\beta}} = \left(\sum_{i=1}^n \underset{\sim}{x}_i\underset{\sim}{x}_i'\right)^{-1} \sum_{i=1}^n \underset{\sim}{x}_i y_i$, where we assume the design matrix is full rank.

3. Consider the residuals, $y_i - \underset{\sim}{x}_i'\hat{\underset{\sim}{\beta}}, i = 1, \ldots, n$. Using the residuals in G, we can obtain the intra-class correlation, $\hat{\rho}$. This can be obtained by pairing the residuals. If $\hat{\rho} < 0$, take $\hat{\rho} = 0$. Also, let S_0^2 denote the sample variance of the residuals in G and S_1^2 the sample variance of the residuals in G^c. Then, $\hat{\rho} = \frac{S_0^2}{S_1^2}$, and if $\hat{\rho} > 1$, take $\hat{\rho} = 1$.

4. Noting that $\text{Var}(y_i) = \{(1-p)\rho + p\}\,\sigma^2, i = 1, \ldots, n$, we have

$$\{(1-\hat{p})\hat{\rho} + \hat{p}\}\,\hat{\sigma^2} = \frac{\sum_{i=1}^n (y_i - \underset{\sim}{x}_i'\underset{\sim}{\beta})^2}{n-c},$$

   where c is the number covariates including the intercept. That is,

$$\hat{\sigma^2} = \frac{\sum_{i=1}^n (y_i - \underset{\sim}{x}_i'\underset{\sim}{\beta})^2}{(n-c)\{(1-\hat{p})\hat{\rho} + \hat{p}\}}.$$

Therefore, the parameters, $p, \underset{\sim}{\beta}, \rho, \sigma^2$, are identifiable. Of course, fitting the Bayesian model with prior,

$$\pi(p, \underset{\sim}{\beta}, \rho, \sigma^2) \propto \frac{1}{\sigma^2}, 0 < p < \frac{1}{2}, 0 < \rho < 1,$$

will provide posterior densities of these parameters, and better posterior summaries.

Appendix D: Propriety of the Joint Posterior Density of the Mixture Model

We consider the Bayesian version of our mixture model,

$$y_i \mid p, \underset{\sim}{\beta}, \sigma^2, \rho \overset{ind}{\sim} (1-p)\text{Normal}\left(\underset{\sim}{x}_i'\underset{\sim}{\beta}, \rho\sigma^2\right) + p\text{Normal}\left(\underset{\sim}{x}_i'\underset{\sim}{\beta}, \sigma^2\right), i = 1, \ldots, n,$$

$$\pi(p, \underset{\sim}{\beta}, \rho, \sigma^2) \propto \frac{1}{\sigma^2}, 0 < p < \frac{1}{2}, 0 < \rho < 1.$$

Without loss of generality, we drop the survey weights. The joint posterior density is

$$\pi(p, \underset{\sim}{\beta}, \rho, \sigma^2 \mid \underset{\sim}{y}) \propto$$

$$\frac{1}{\sigma^2}\prod_{i=1}^{n}\left\{(1-p)\text{Normal}\left(\underset{\sim}{x}_i'\underset{\sim}{\beta}, \rho\sigma^2\right) + p\text{Normal}\left(\underset{\sim}{x}_i'\underset{\sim}{\beta}, \sigma^2\right)\right\}, 0 < p < \frac{1}{2}, 0 < \rho < 1.$$

It is easy to show that the joint posterior density is proper.

For this purpose, we introduce independent binary latent variables, $z_i = 0, 1, i = 1, \ldots, n$. Then, $(y_i, z_i), i = 1, \ldots, n$, are independent, and the joint posterior density is

$$\pi(\underset{\sim}{z}, p, \underset{\sim}{\beta}, \rho, \sigma^2 \mid \underset{\sim}{y}) \propto$$

$$\frac{1}{\sigma^2} \times \prod_{i=1}^{n}\left\{(1-p)\text{Normal}\left(\underset{\sim}{x}_i'\underset{\sim}{\beta}, \rho\sigma^2\right)\right\}^{1-z_i}\left\{p\text{Normal}\left(\underset{\sim}{x}_i'\underset{\sim}{\beta}, \sigma^2\right)\right\}^{z_i}, 0 < p < \frac{1}{2}, 0 < \rho < 1.$$

Assuming that the design matrix is full rank, it is easy to integrate out $(p, \underset{\sim}{\beta}, \sigma^2)$ to get $\pi(\underset{\sim}{z}, \rho \mid \underset{\sim}{y})$.

We can now sum $\pi(\underset{\sim}{z}, \rho \mid \underset{\sim}{y})$ over all z_i (n summations) to get $\pi(\rho \mid \underset{\sim}{y})$. Then, finally assuming $\epsilon \leq \rho \leq 1 - \epsilon$, where ϵ is a small positive real number, the integral over ρ is finite. Therefore, $\pi(p, \underset{\sim}{\beta}, \rho, \sigma^2 \mid \underset{\sim}{y})$ is proper.

Appendix E: Summary results of population parameters

Table 3: Summaries of the parameters of non-robust population model.

	Summaries	β_0	β_1	β_2	β_3	σ^2
TR	ESS	1000	1000	1000	1000	1000
	p-value	0.140	0.297	0.633	0.053	0.521
UNP	ESS	1000	1000	1000	1000	1000
	p-value	0.649	0.566	0.667	0.681	0.556
UNNP	ESS	1000	1000	1000	1000	1000
	p-value	0.755	0.605	0.933	0.639	0.713

Note: Summaries are Effective Sample Size (ESS) and p-value of Geweke test.

Table 4: Posterior summaries of non-robust population model.

	Summaries	β_0	β_1	β_2	β_3	σ^2
TR	PM	27.613	-0.012	0.647	0.482	15.079
	PSD	0.451	0.007	0.453	0.261	0.549
	NSE	0.0002	0.0142	0.008	0.014	0.017
	PCV	0.016	0.583	0.7	0.541	0.036
	95% HPDI	(26.773,28.507)	(-0.025,0.001)	(-0.230,1.529)	(0.003,1.011)	(14.06,16.14)
UNP	PM	28.909	-0.029	0.494	0.429	15.278
	PSD	2.609	0.039	0.874	0.345	7.598
	NSE	0.001	0.020	0.010	0.066	0.221
	PCV	0.090	1.344	1.769	0.804	0.497
	95% HPDI	(25.725,35.326)	(-0.120,0.026)	(-0.927,1.661)	(-0.277,0.990)	(9.277,22.660)
UNNP	PM	28.473	-0.022	0.498	0.467	16.281
	PSD	2.098	0.032	0.633	0.326	6.991
	NSE	0.001	0.020	0.010	0.066	0.221
	PCV	0.074	1.455	1.271	0.698	0.429
	95% HPDI	(25.659,33.670)	(-0.099,0.250)	(-0.743,1.561)	(-0.191,1.072)	(10.689,22.137)

Note: Summaries are PM = posterior mean, PSD = posterior standard deviation, NSE = numeric standard error, PCV = posterior coefficient of variation, HPDI = highest posterior density interval.

Table 5: Summaries of the parameters of robust population model.

	Summaries	β_0	β_1	β_2	β_3	σ^2	ρ	p
TR-R	ESS	1000	1000	1000	1000	1000	1000	1000
	p-value	0.902	0.862	0.218	0.557	0.413	0.957	0.053
UNP-R	ESS	1000	1000	1000	1000	1000	1000	1000
	p-value	0.487	0.575	0.402	0.312	0.858	0.856	0.684
UNNP-R	ESS	1000	1000	1000	1000	1000	1000	1000
	p-value	0.736	0.825	0.556	0.113	0.737	0.822	0.157

Note: Summaries are Effective Sample Size (ESS) and p-value of Geweke test.

Table 6: Posterior summaries of robust population model.

	Summaries	β_0	β_1	β_2	β_3	σ^2	ρ	p
TR-R	PM	26.940	-0.001	0.030	-0.024	53.19	0.167	0.159
	PSD	0.433	0.007	0.400	0.260	12.334	0.033	0.040
	NSE	0.013	0.0002	0.012	0.008	0.391	0.001	1.971
	PCV	0.016	7.00	13.3	10.8	0.231	0.197	0.251
	95% HPDI	(26.095,27.809)	(-0.015,0.012)	(-0.773,0.790)	(-0.515,0.495)	(37.191,83.829)	(0.102,0.236)	(0.084,0.236)
UNP-R	PM	27.425	-0.008	0.171	0.034	35.972	0.137	0.313
	PSD	1.707	0.025	0.688	0.314	16.609	0.048	0.155
	NSE	0.053	0.0007	0.021	0.009	0.513	0.001	0.004
	PCV	0.062	3.125	5.70	9.235	0.462	0.35	0.495
	95% HPDI	(25.284,31.656)	(-0.070,0.024)	(-0.932,1.366)	(-0.574,0.631)	(6.392,68.697)	(0.038,0.222)	(0.031,0.499)
UNNP-R	PM	27.209	-0.005	0.228	0.031	37.399	0.148	0.305
	PSD	1.564	0.023	0.609	0.292	15.662	0.048	0.140
	NSE	0.048	0.0007	0.019	0.009	0.498	0.001	0.004
	PCV	0.057	4.60	0.57	0.119	0.418	0.324	0.459
	95% HPDI	(25.239,30.635)	(-0.057,0.027)	(-0.866,1.377)	(-0.551,0.636)	(9.686,68.290)	(0.048,0.237)	(0.051,0.498)

Note: Summaries are PM = posterior mean, PSD = posterior standard deviation, NSE = numeric standard error, PCV = posterior coefficient of variation, HPDI = highest posterior density interval.

Appendix F: Boxplots for adjusted weights of all units in non-probability sample

In Figures 1 and 2, we show posterior densities of $\bar{Y}$ under the six models.

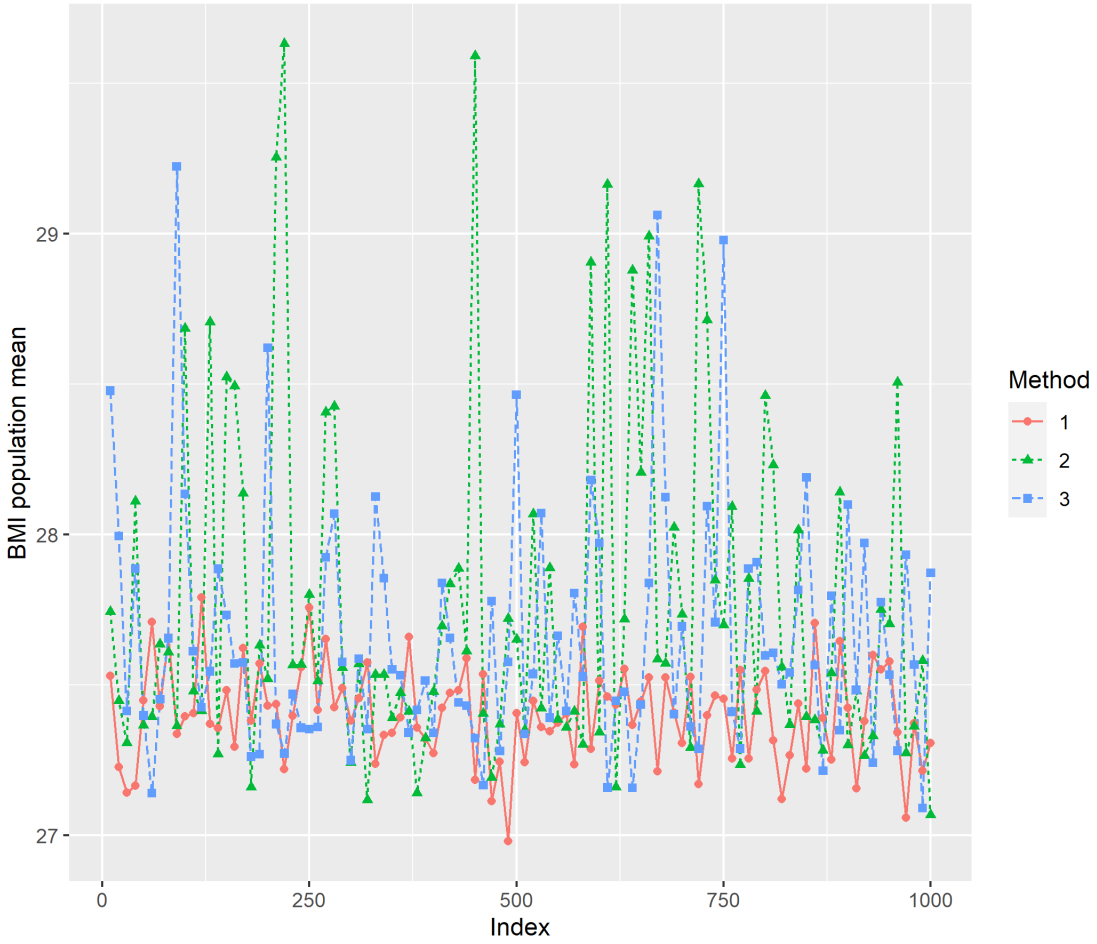


Figure 4: Posterior distributions of $\bar{Y}$ for TR(1), UNP(2) and UNNP(3).

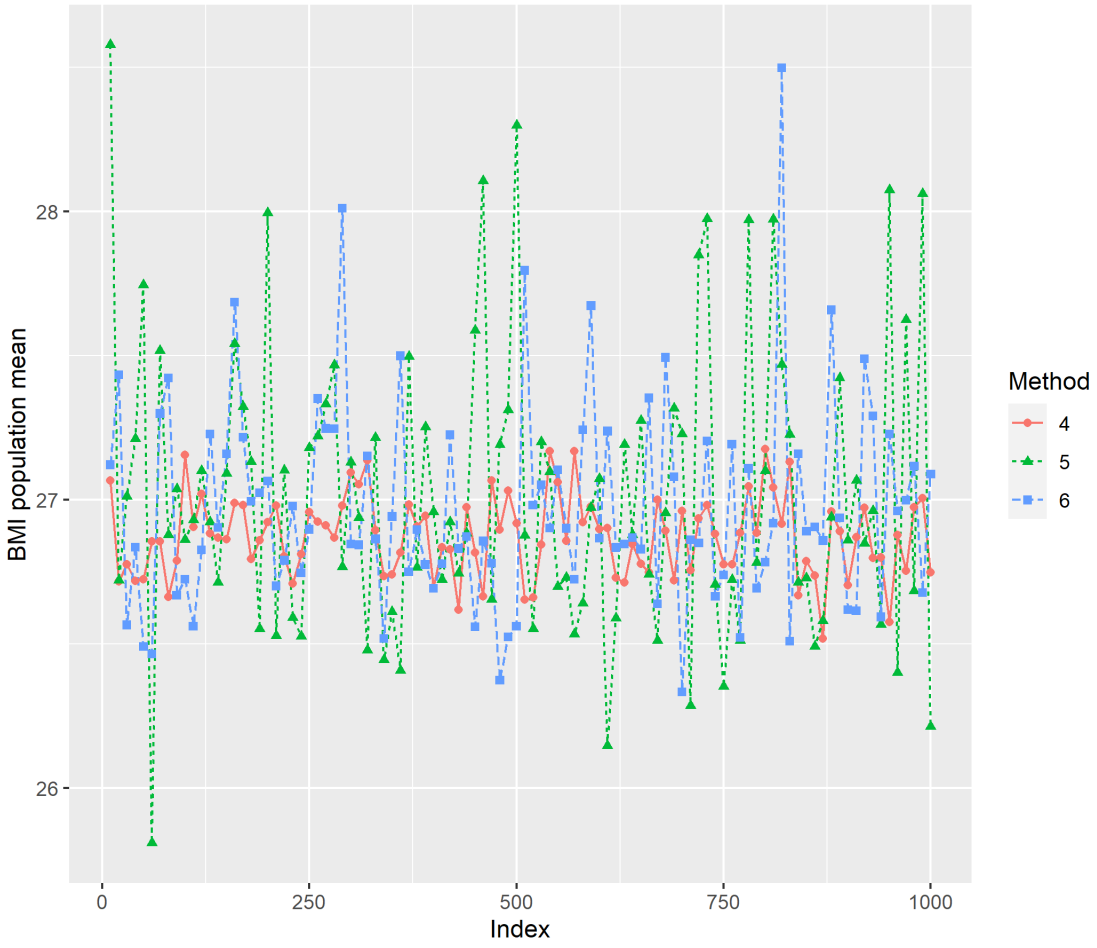


Figure 5: Posterior distributions of $\bar{Y}$ for TR-R(4), UNP-R(5) and UNNP-R(6).

Appendix G: Boxplots for adjusted weights of all units in non-probability sample

In Appendix G, we randomly select 500 units from the remaining observations, excluding the first 100 units in the non-probability sample. The boxplots represent the posterior density of each adjusted weight, obtained from the 1,000 iterates of the Gibbs sampler. We can see that some adjusted weights are very small while others are very large. Also, some adjusted weights are much more variable than others.

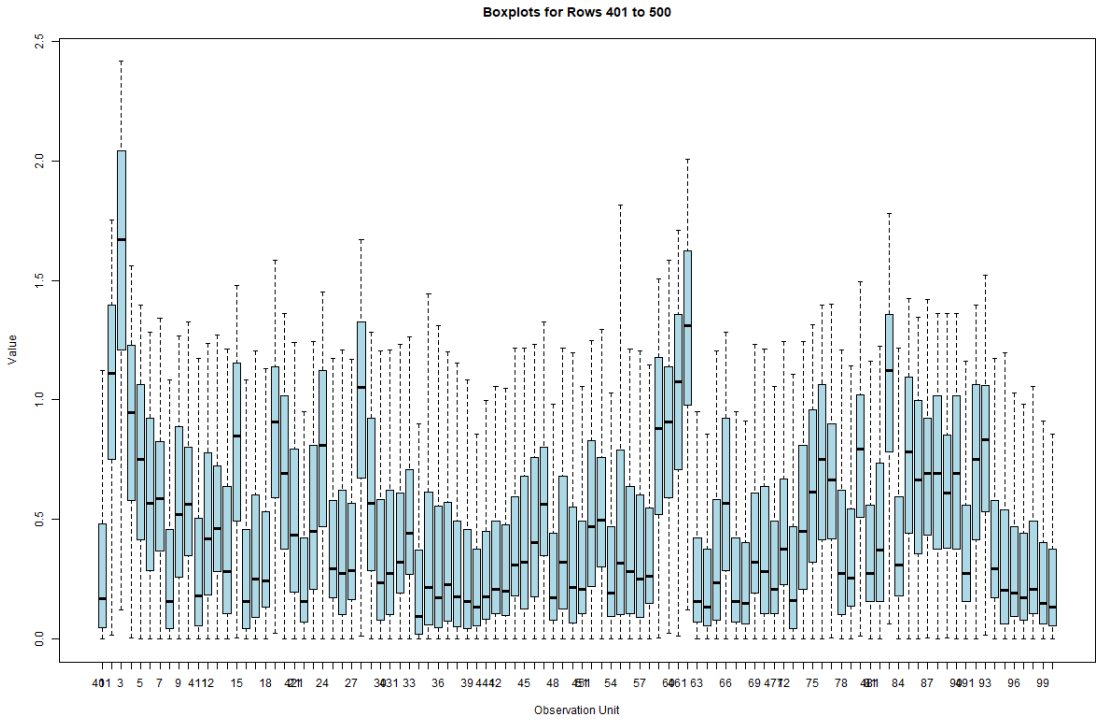


Figure 6: Boxplots for units in non-probability sample.

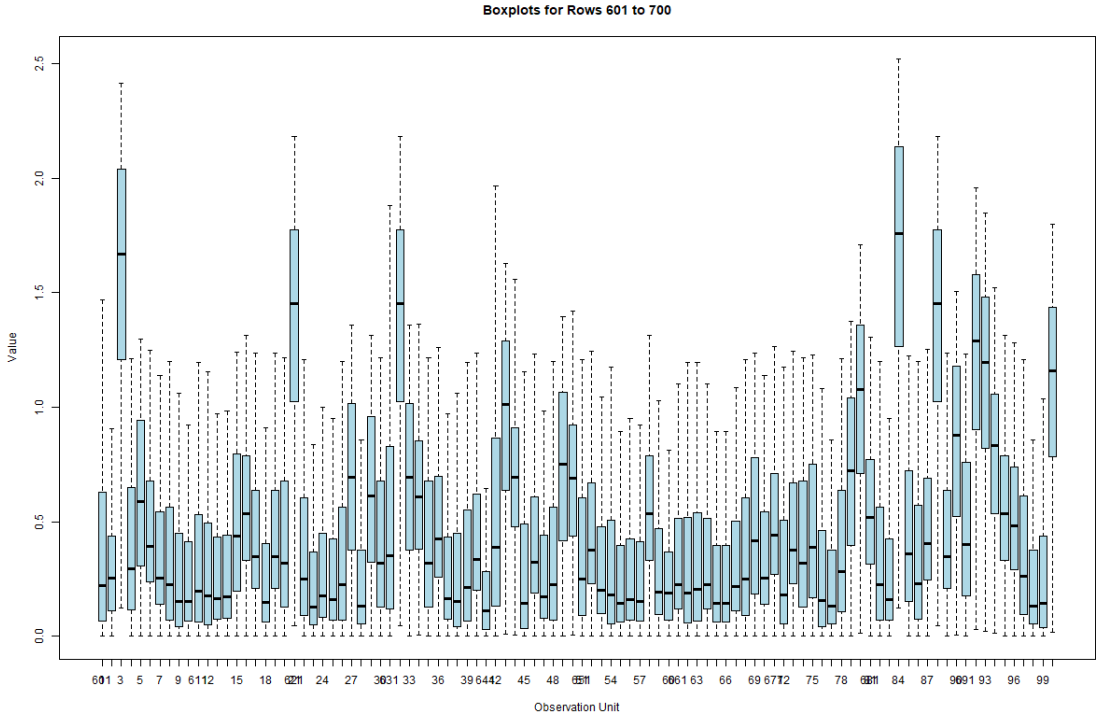


Figure 7: Boxplots for units in non-probability sample.

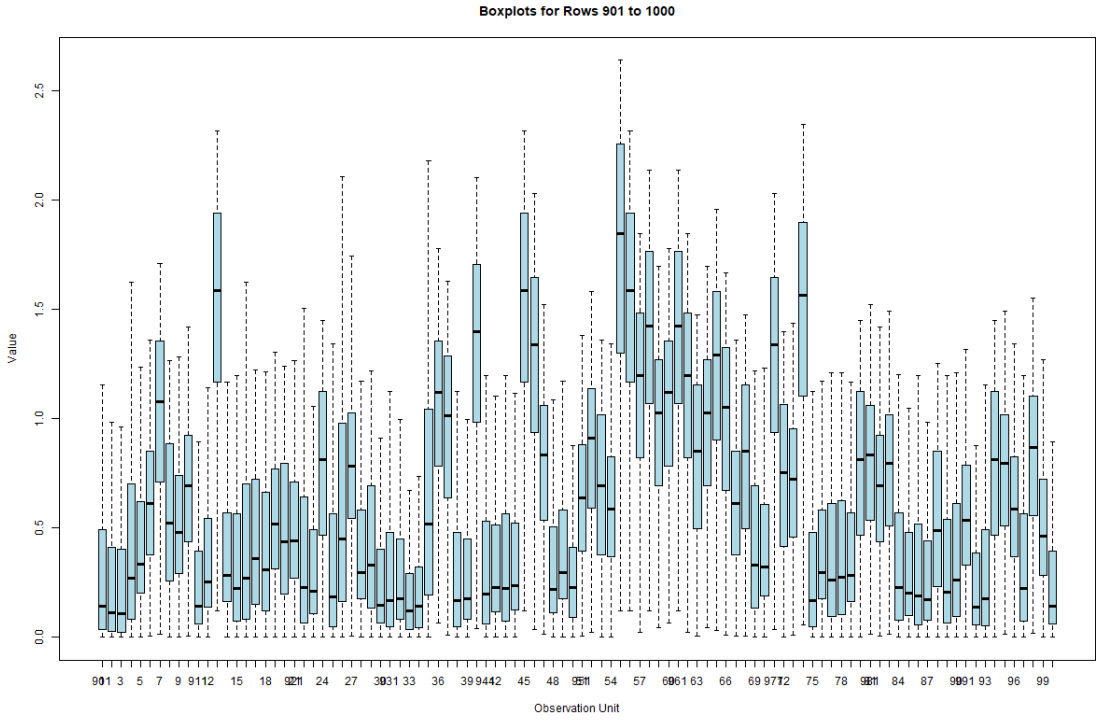


Figure 8: Boxplots for units in non-probability sample.

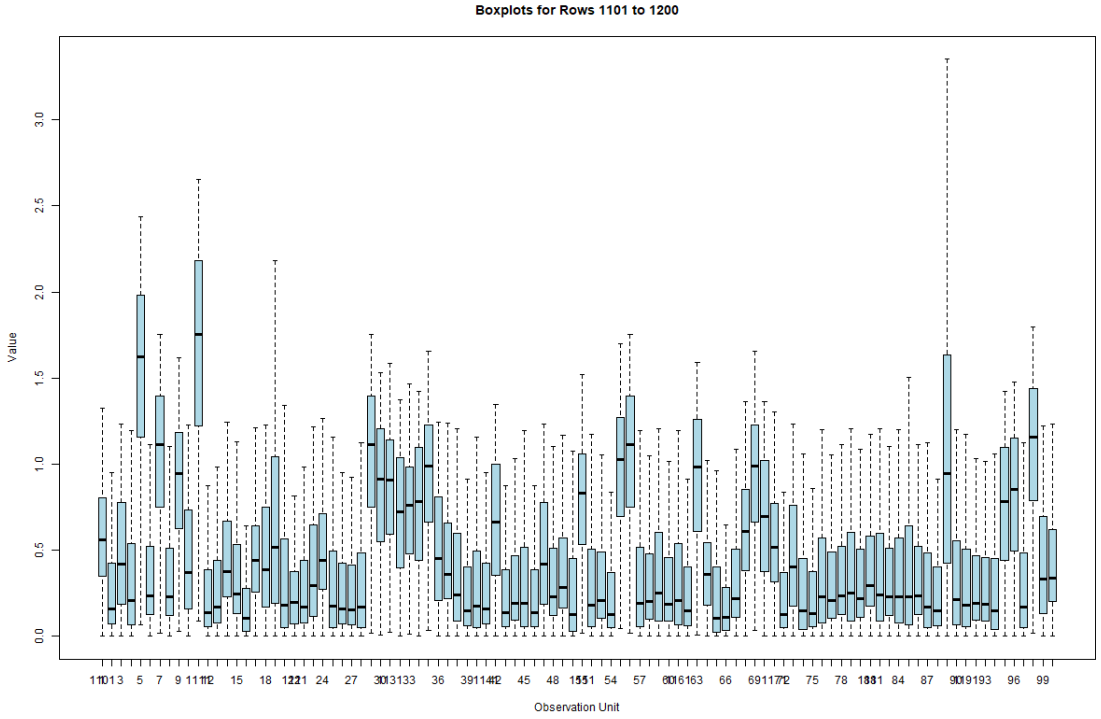


Figure 9: Boxplots for units in non-probability sample.

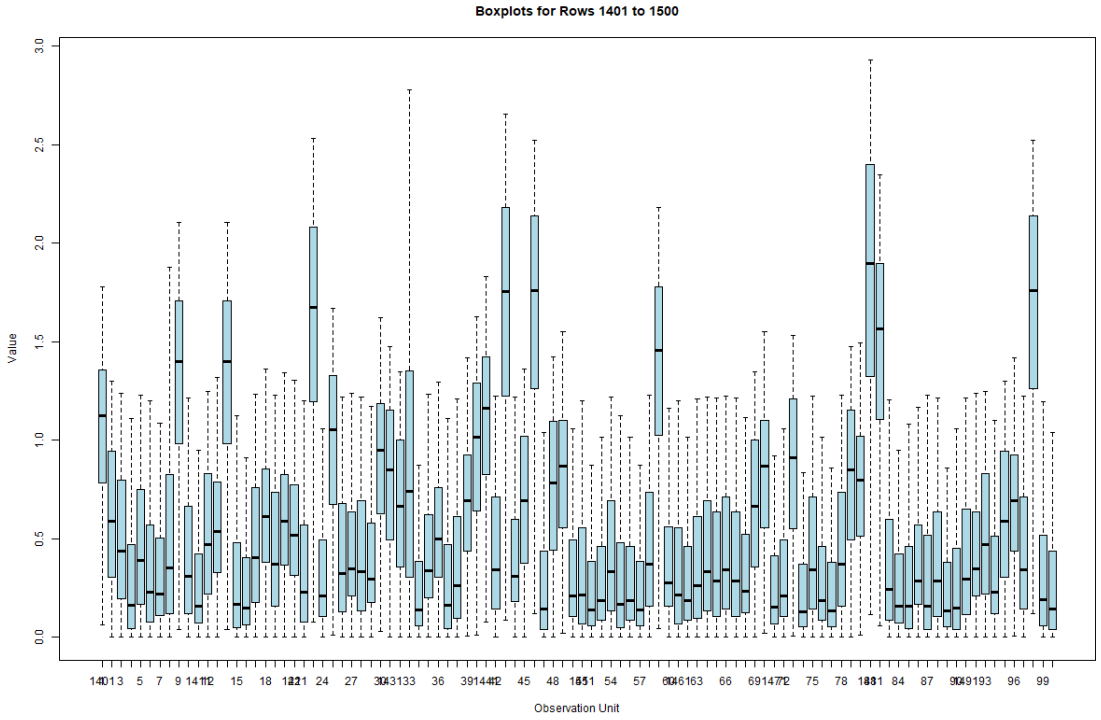


Figure 10: Boxplots for units in non-probability sample.